

The Over-Dispersed Skellam (ODS) Reserving Model

A Signed Stochastic Framework for Incurred Chain-Ladder Uncertainty

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Abstract

Case-reserve releases and recoveries sometimes produce negative incremental incurred development, motivating predictive models whose distribution for aggregate future incurred development can assign positive probability to both positive and negative outcomes. We develop an over-dispersed Skellam (ODS) predictive layer around a selected incurred chain-ladder (CL) completion. Each cell is the difference of independent negative-binomial components. Their future totals are independent negative-binomial variables with a common success probability. The aggregate has the mean and variance of a Skellam difference with its variance multiplied by the dispersion parameter. The fitted, one-sided layer preserves the selected CL mean in every future cell and has an over-dispersed Poisson (ODP) mean–variance relation when all increments are non-negative. When future cells include both fitted signs, the aggregate has support on both sides of zero. An individual cell’s fitted sign is fixed by development age. We examine Mack-MSEP and absolute-value Pearson dispersion calibrations. Mack-MSEP calibration matches the fitted aggregate process variance to the Mack MSEP target, matching one aggregate moment rather than the full predictive distribution. In a 30-by-30 synthetic experiment, empirical 95% interval coverage was 88.5% for the process-only benchmark, 95.8% for Mack-MSEP, and 84.0% for the exploratory absolute-residual scale. These plug-in intervals do not explicitly propagate parameter-estimation uncertainty, and the coverage comparison is specific to the stated design.

Keywords: claims reserving, incurred triangles, chain-ladder, mean squared error of prediction, over-dispersed Skellam model, prediction uncertainty, negative incremental claims.

1 Introduction

Stochastic reserving methods supplement a deterministic reserve estimate with a predictive distribution (England and Verrall, 2002, 2006; Wüthrich and Merz, 2008; Scarth *et al.*, 2020). For incurred triangles, case-reserve reassessments, recoveries, and settlements below previously established case estimates can produce negative incremental incurred development. We study the aggregate future incurred development R , conditional on the observed triangle and centred on a selected incurred chain-ladder (CL) completion. Such signed development motivates a predictive distribution for aggregate future incurred development that can assign probability to either side of zero when future cells have both positive and negative fitted means. Mapping this quantity to payment projections or reported but not settled (RBNS) and incurred but not reported (IBNR) amounts requires additional claims and accounting information.

The over-dispersed Skellam (ODS) model presented here is a predictive layer around a selected CL completion. A cell is the difference of independent negative-binomial components. Its mean is μ_{ij} and its variance is φ times the sum of its component means, where $\varphi > 1$ is the dispersion parameter. For fixed component means, let M_+ and M_- be their positive- and negative-component totals over future cells. Then

$$\mathbb{E}[R] = M_+ - M_-, \quad \text{Var}(R) = \varphi(M_+ + M_-).$$

Thus, the aggregate shares the mean of a classical Skellam difference with component means M_+ and M_- , with variance inflated by φ . It is a difference of independent negative-binomial totals, not a classical Skellam variable. The fitted layer imposes a one-sided sign restriction at each development age, giving cell variance $\varphi|\mu_{ij}|$, but the population specification allows both components to be positive. Thus, two-sided aggregate support does not imply that an individual fitted cell can reverse its age's sign. A zero-mean fitted cell is deterministic. The Mack-MSEP calibration matches the fitted aggregate process variance to the Mack-MSEP target, a match of one aggregate second moment, while the Pearson-type calibration uses observed-cell residuals.

Negative incremental claims have been considered in chain-ladder and linear models (Verrall and Li, 1993), and negative runoff-triangle values in a three-parameter log-normal approach (de Alba, 2006). Related stochastic chain-ladder GLM work is described by Renshaw and Verrall (1998). This paper adds an explicit signed discrete predictive distribution around a selected incurred CL mean surface, together with aggregate dispersion calibrations. ODS assumes conditional independence of signed negative-binomial cells, whereas Mack specifies accident-year independence and conditional moments for cumulative development paths (Mack, 1994). A recent stochastic lift of the generalized Cape Cod (GCC) method derives an analytical MSEP for a different mean estimator (Richman and Wüthrich, 2026). The present focus is the signed discrete predictive distribution for incurred increments.

The contribution is an actuarial construction: a predictive layer anchored to a selected incurred CL mean surface, with an aggregate Mack-MSEP variance match and an exploratory observed-cell absolute-

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residual scale. A synthetic study illustrates the resulting interval behaviour. Portfolio-specific validation has also been undertaken through actuarial-function work, but is not reported here. The reproducible numerical evidence in this paper is limited to the synthetic design.

2 The model

We define the deterministic incurred CL anchor first and then place the ODS predictive layer around it. The notation also records the observed upper triangle and the future lower triangle.

2.1 Triangle and incurred CL anchor

Let X_{ij} be incremental incurred development for accident period $i = 1, \dots, I$ and development period $j = 0, \dots, J$. For the CL calculations below take $J \geq 1$ and $I \geq J + 1$. At valuation time I , define

$$\begin{aligned}\mathcal{I}_I &= \{(i, j) : 1 \leq i \leq I, 0 \leq j \leq J, i + j \leq I\}, \\ \mathcal{L}_I &= \{(i, j) : 1 \leq i \leq I, 0 \leq j \leq J, i + j > I\}.\end{aligned}$$

The set $\mathcal{I}_I$ is the observed upper triangle. The set $\mathcal{L}_I$ is the future lower triangle. For each accident period define the last observed development period

$$j_i^* := \min\{J, I - i\}, \quad i = 1, \dots, I.$$

Note that in the square-design special case, $j_i^* = I - i$. The observed data are $\mathcal{D}_I = (X_{ij})_{(i,j) \in \mathcal{I}_I}$ and $\mathcal{F}_I = \sigma(\mathcal{D}_I)$. All fitted predictive quantities below are conditional on $\mathcal{F}_I$. The target is aggregate future incurred development,

$$R = \sum_{(i,j) \in \mathcal{L}_I} X_{ij}.$$

For $i = 1, \dots, I$ and $j = 0, \dots, J$, define cumulative incurred development by

$$C_{ij} = \sum_{k=0}^j X_{ik}, \quad C_{i,-1} = 0.$$

The selected volume-weighted incurred CL factor is

$$\hat{f}_j = \frac{\sum_{i=1}^{I-j-1} C_{i,j+1}}{\sum_{i=1}^{I-j-1} C_{ij}}, \quad j = 0, \dots, J-1,$$

provided the denominator is positive and finite. Note that negative incrementals are allowed. The standard Mack calculation requires positive cumulative denominators and positive selected factors. For $i = 1, \dots, I$ and $j = 0, \dots, J$, define the completed cumulative values by

$$\hat{C}_{ij}^{\text{CL}} = \begin{cases} C_{ij}, & (i, j) \in \mathcal{I}_I, \\ \hat{f}_{j-1} \hat{C}_{i,j-1}^{\text{CL}}, & (i, j) \in \mathcal{L}_I. \end{cases}$$

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For $i = 1, \dots, I$ and $j = 0, \dots, J$, its incremental means are

$$\hat{\mu}_{ij}^{\text{CL}} := \hat{C}_{ij}^{\text{CL}} - \hat{C}_{i,j-1}^{\text{CL}}, \quad \hat{C}_{i,-1}^{\text{CL}} = 0.$$

The selected CL reserve sums these projected future increments. By telescoping within each accident period, it is equivalently the projected ultimate less the latest observed cumulative:

$$\hat{R}^{\text{CL}} := \sum_{(i,j) \in \mathcal{L}_I} \hat{\mu}_{ij}^{\text{CL}} = \sum_{i=1}^I \left(\hat{C}_{iJ}^{\text{CL}} - C_{i,j_i^*} \right).$$

2.2 Mack benchmark and calibration target

The Mack CL model is used here as the deterministic incurred-CL benchmark and as an aggregate calibration target. We state its three model assumptions separately, following [Mack \(1993\)](#):

Assumption 1 (Mack CL assumptions). For $i = 1, \dots, I$ and $j = 0, \dots, J - 1$:

(MCL1) The accident-period cumulative paths $(C_{i0}, \dots, C_{iJ})$ are mutually independent across i .

(MCL2) There exist development factors $f_j > 0$ such that

$$\mathbb{E}[C_{i,j+1} \mid C_{i0}, \dots, C_{ij}] = f_j C_{ij}.$$

(MCL3) There exist variance parameters $\sigma_j^2 > 0$ such that

$$\text{Var}(C_{i,j+1} \mid C_{i0}, \dots, C_{ij}) = \sigma_j^2 C_{ij}.$$

MCL1–MCL3 define the benchmark at accident-period level. MCL1 separates accident periods, while MCL2–MCL3 specify their conditional first two moments. The selected CL completion supplies the ODS mean anchor, and the Mack MSEP supplies the aggregate calibration target.

Mack regularity conditions. For the CL and Mack formulas below, observed cumulative values are non-negative and each selected development factor has at least one age-to-age pair. Whenever $C_{i,j+1}/C_{ij}$ is used, require $C_{ij} > 0$ for $i = 1, \dots, I - j - 1$ and $j = 0, \dots, J - 1$. The displayed variance formulas are not defined for a square triangle with $J < 3$; no alternative terminal-age estimator is specified here. All denominators and variance estimates used below must be positive and finite. For $j = 0, \dots, J - 2$, use

$$\hat{\sigma}_j^2 = \frac{1}{I - j - 2} \sum_{i=1}^{I-j-1} C_{ij} \left(\frac{C_{i,j+1}}{C_{ij}} - \hat{f}_j \right)^2.$$

In the square design used in the numerical study, $I = J + 1$. The terminal age $j = J - 1$ then has exactly one observed link ratio. In this one-ratio case, use the standard Mack extrapolation

$$\hat{\sigma}_{J-1}^2 = \min \left\{ \hat{\sigma}_{J-2}^2, \hat{\sigma}_{J-3}^2, \frac{(\hat{\sigma}_{J-2}^2)^2}{\hat{\sigma}_{J-3}^2} \right\},$$

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when $I = J + 1$, $J \geq 3$, and the preceding estimates are positive (Mack, 1993, 1999). For a rectangular array with $I \geq J + 2$, the terminal age has at least two observed link ratios, and its variance coefficient is instead estimated directly by

$$\hat{\sigma}_{J-1}^2 = \frac{1}{I - J - 1} \sum_{i=1}^{I-J} C_{i,J-1} \left(\frac{C_{iJ}}{C_{i,J-1}} - \hat{f}_{J-1} \right)^2.$$

This distinguishes the one-ratio square case from the rectangular case. The aggregate Mack MSEP used for calibration is

$$\widehat{\text{MSEP}}_{\text{Mack}}(R) = \underbrace{\sum_{i=1}^I (\hat{C}_{ij}^{\text{CL}})^2 \sum_{j=j_i^*}^{J-1} \frac{\hat{\sigma}_j^2}{(\hat{f}_j)^2 \hat{C}_{ij}^{\text{CL}}}}_{\text{process uncertainty}} + \underbrace{\sum_{j=0}^{J-1} \frac{\hat{\sigma}_j^2}{(\hat{f}_j)^2} \left(\sum_{\ell=1}^{I-j-1} C_{\ell j} \right) \left(\sum_{i=I-j}^I \hat{C}_{ij}^{\text{CL}} \right)^2}_{\text{estimation uncertainty}}. \quad (1)$$

This estimated aggregate MSEP is the benchmark for the error $R - \hat{R}^{\text{CL}}$. Its first term is estimated future process variation under Mack and its second term accounts for estimating the CL factors under Mack. Neither term is separately a distribution for that error. The ODS distribution treats its fitted parameters as fixed plug-in values.

2.3 The ODS model

The ODS layer retains the accident-period/development-period decomposition and models each incremental cell using a sign and a nonnegative magnitude. Let

$$\theta := (\alpha_1, \dots, \alpha_I, \beta_0^+, \dots, \beta_J^+, \beta_0^-, \dots, \beta_J^-).$$

Define the cell mean by

$$\mu_{ij} := \alpha_i(\beta_j^+ - \beta_j^-).$$

The ODS distribution has integer support, so observed incremental amounts and simulated cell realisations must be integer multiples of a stated monetary unit. Fitted means and component intensities may be non-integer in that unit and their real-valued negative-binomial shapes need not be rounded. Changing that unit changes the numerical cell means and the dispersion: for a unit multiplied by $c > 0$, matching the same monetary mean and variance requires φ to be divided by c at the moment level. This does not make the discrete distributions invariant to rescaling or rounding, and the new scale must still satisfy $\varphi > 1$. We write $\text{NB}(r, p)$ for the negative-binomial distribution with real-valued size $r > 0$ and success probability $p \in (0, 1)$. Its probability mass function is

$$\mathbb{P}(N = n) = \frac{\Gamma(n+r)}{\Gamma(r)n!} p^r (1-p)^n, \quad n = 0, 1, 2, \dots,$$

and it has mean and variance given by

$$\mathbb{E}[N] = \frac{r(1-p)}{p}, \quad \text{Var}(N) = \frac{r(1-p)}{p^2}.$$

The ODS population and sampling assumptions are as follows:

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Assumption 2 (ODS assumptions).

(ODS1) The triangle dimensions are finite, cells are integer-valued, $\varphi > 1$, $\alpha_i > 0$ for $i = 1, \dots, I$, and $\beta_j^\pm \geq 0$ for $j = 0, \dots, J$.

(ODS2) All component variables are mutually independent across cells and signs. For each cell define non-negative components Y_{ij}^+ and Y_{ij}^- with

$$Y_{ij}^+ \sim \text{NB}\left(\frac{\alpha_i \beta_j^+}{\varphi - 1}, \frac{1}{\varphi}\right), \quad Y_{ij}^- \sim \text{NB}\left(\frac{\alpha_i \beta_j^-}{\varphi - 1}, \frac{1}{\varphi}\right).$$

If $\alpha_i \beta_j^+ = 0$, set $Y_{ij}^+ = 0$ almost surely. If $\alpha_i \beta_j^- = 0$, set $Y_{ij}^- = 0$ almost surely. A zero net mean $\mu_{ij} = 0$ need not give a degenerate cell if both components are positive.

Model 1 (The ODS reserving model). For $i = 1, \dots, I$ and $j = 0, \dots, J$, define the incremental cells by

$$X_{ij} := Y_{ij}^+ - Y_{ij}^-,$$

where (Y_{ij}^+, Y_{ij}^-) is specified in ODS2 of assumption 2.

The cell mean is μ_{ij} and the cell variance is $\text{Var}(X_{ij}) = \varphi \alpha_i (\beta_j^+ + \beta_j^-)$. Because $\mu_{ij} = \alpha_i (\beta_j^+ - \beta_j^-)$, for any $c > 0$ the transformation $(\alpha_i, \beta_j^\pm) \mapsto (c\alpha_i, \beta_j^\pm/c)$ leaves every cell mean, and hence every cell distribution, unchanged. The factors are therefore not separately identified. We impose the fitted normalisation below as a scale convention. It leaves the fitted cell means unchanged and sets $\hat{\alpha}_{iI}$ equal to the fitted CL ultimate $\hat{U}_I$:

$$\sum_{j=0}^J (\hat{\beta}_{jI}^+ - \hat{\beta}_{jI}^-) = 1.$$

Remark 1 (Mack and ODS dependence assumptions). *MCL1 allows dependence among incremental cells within an accident period, whereas ODS2 implies independence of the cells conditional on the fixed parameters. Calendar-year effects act along triangle diagonals and can induce dependence across accident periods, adding a natural extension of this development-level specification. The present ODS distribution focuses on cellwise variation while retaining the ability to represent negative incrementals in incurred data.*

2.4 Fitted ODS construction and dispersion calibration

For the fitted predictive layer, all quantities defined below are deterministic functions of $\mathcal{D}_I$ and hence $\mathcal{F}_I$ -measurable. We use φ for the population dispersion parameter and write the two data-driven calibrations as $\hat{\varphi}_{\text{Mack}}$ and $\hat{\varphi}_{\text{Pearson}}$. Thus, $\hat{\varphi}_I$ is the selected value of one of these calibrations,

$$\hat{\varphi}_I \in \{\hat{\varphi}_{\text{Mack}}, \hat{\varphi}_{\text{Pearson}}\}, \quad \hat{\varphi}_I > 1.$$

Let $\hat{\mathbb{P}}_I^{\text{ODS}}$ denote the fitted plug-in distribution for the future vector $(X_{ij})_{(i,j) \in \mathcal{L}_I}$ obtained by replacing (θ, φ) in model 1 by the fitted parameters defined below and $\hat{\varphi}_I$. The observed cells are held fixed at $\mathcal{D}_I$. We write $\hat{\mathbb{E}}_I$ and $\hat{\text{Var}}_I$ for moments under this distribution. It is a plug-in approximation to the

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conditional distribution under the original model and uses the signed negative-binomial construction in model 1. Define the CL ultimates by $\hat{U}_i := \hat{C}_{ij}^{\text{CL}}$ for $i = 1, \dots, I$. The CL-to-ODS mapping is well-defined when

$$0 < \hat{U}_i < \infty, \quad 0 < |\hat{f}_j| < \infty, \quad i = 1, \dots, I, \quad j = 0, \dots, J-1.$$

Define the development shares by

$$\hat{\gamma}_0 := \frac{1}{\prod_{k=0}^{J-1} \hat{f}_k}, \quad \hat{\gamma}_j := \frac{\hat{f}_{j-1} - 1}{\prod_{k=j-1}^{J-1} \hat{f}_k}, \quad j = 1, \dots, J.$$

For the Mack MSEP calibration, the Mack requirements impose $\hat{f}_j > 0$ and positive cumulative denominators. In addition, at least one future cell must have positive component intensity. Otherwise the Mack MSEP denominator is zero. The nonzero-factor condition for the CL-to-ODS mapping is imposed in addition to the Mack requirements.

Here and below, an empty product equals one and an empty sum equals zero. For every future cell $(i, j) \in \mathcal{L}_I$, so that $j \geq j_i^* + 1$, the completed CL recursion gives

$$\hat{C}_{ij}^{\text{CL}} = \frac{\hat{U}_i}{\prod_{k=j}^{J-1} \hat{f}_k}.$$

Consequently, $\hat{U}_i \hat{\gamma}_j$ is the completed CL increment at age j (whenever that age is used in the future triangle). The fitted ODS parameters constructed from the CL increments are defined by

$$\hat{\alpha}_{iI} := \hat{U}_i, \quad \hat{\beta}_{jI}^+ := \max(\hat{\gamma}_j, 0), \quad \hat{\beta}_{jI}^- := \max(-\hat{\gamma}_j, 0).$$

Here $i = 1, \dots, I$ for $\hat{U}_i$ and $\hat{\alpha}_{iI}$, and $j = 0, \dots, J$ for $\hat{\gamma}_j$ and $\hat{\beta}_{jI}^\pm$. By construction, the fitted parameters satisfy

$$\hat{\beta}_{jI}^+ \hat{\beta}_{jI}^- = 0, \quad j = 0, \dots, J. \quad (2)$$

Thus, at most one signed component is nonzero at each fitted development period. The population parameters in ODS1 of assumption 2 remain unrestricted. Consequently,

$$\hat{\mu}_{ij}^{\text{ODS}} := \hat{\alpha}_{iI} (\hat{\beta}_{jI}^+ - \hat{\beta}_{jI}^-) = \hat{U}_i \hat{\gamma}_j.$$

Under $\hat{\mathbb{P}}_I^{\text{ODS}}$, for each future cell with $\hat{\mu}_{ij}^{\text{ODS}} \neq 0$, the signed-magnitude distribution is

$$X_{ij} \stackrel{d}{=} \text{sign}(\hat{\mu}_{ij}^{\text{ODS}}) N_{ij}, \quad N_{ij} \sim \text{NB} \left(\frac{|\hat{\mu}_{ij}^{\text{ODS}}|}{\hat{\phi}_I - 1}, \frac{1}{\hat{\phi}_I} \right).$$

If $\hat{\mu}_{ij}^{\text{ODS}} = 0$, then $X_{ij} = 0$ almost surely. Since $\sum_{j=0}^J \hat{\gamma}_j = 1$, this normalisation makes $\hat{\alpha}_{iI}$ equal to the completed incurred CL ultimate. The completed CL recursion implies that $\hat{\mu}_{ij}^{\text{ODS}} = \hat{\mu}_{ij}^{\text{CL}}$ for future cells

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$(i, j) \in \mathcal{L}_I$. Thus the fitted ODS layer reproduces each future CL cell mean. To make the accident-period implication explicit, define the random completed ultimate for each accident period $i = 1, \dots, I$ by

$$U_i^{\text{ODS}} := C_{i,j_i^*} + \sum_{j=j_i^*+1}^J X_{ij}, \quad \hat{U}_i^{\text{CL}} := \hat{C}_{ij}^{\text{CL}}.$$

Then, conditional on the observed triangle,

$$\hat{\mathbb{E}}_I[U_i^{\text{ODS}}] = C_{i,j_i^*} + \sum_{j=j_i^*+1}^J \hat{\mu}_{ij}^{\text{CL}} = \hat{U}_i^{\text{CL}}, \quad \hat{\mathbb{E}}_I[R] = \hat{R}^{\text{CL}}.$$

The fitted ODS process variance is derived in section 3.

For the Pearson-type calibration, define the fitted observed-cell increment for $(i, j) \in \mathcal{I}_I$ with $j \geq 1$ by

$$\hat{\mu}_{ij}^{\text{P}} := (\hat{f}_{j-1} - 1)C_{i,j-1}.$$

There is no fitted observed-cell increment at $j = 0$: the first column is conditioned upon. This differs from $\hat{\mu}_{ij}^{\text{CL}}$, which equals the observed X_{ij} in the upper triangle. The future ODS distribution uses the CL completion, not these observed-cell residual means. The Mack MSEP in eq. (1) supplies an aggregate target. Its parameter-estimation component stays within the Mack target, while the ODS calibration adds process variation to the fitted plug-in distribution. We calibrate the fitted ODS process-variance scale using the future component-intensity scale:

$$\hat{\phi}_{\text{Mack}} = \frac{\widehat{\text{MSEP}}_{\text{Mack}}(R)}{\sum_{(i,j) \in \mathcal{L}_I} \hat{\alpha}_{iI} (\hat{\beta}_{jI}^+ + \hat{\beta}_{jI}^-)}, \quad (3)$$

provided the denominator is positive and the resulting calibration satisfies $\hat{\phi}_{\text{Mack}} > 1$. Under the fitted one-sided sign structure in eq. (2), the denominator equals $\sum_{(i,j) \in \mathcal{L}_I} |\hat{\mu}_{ij}^{\text{CL}}|$. The coefficient is solely an aggregate calibration for the fitted plug-in distribution. Under the fitted ODS distribution and the one-sided development-period restriction, this choice gives

$$\widehat{\text{Var}}_{I,\text{proc}}(R) = \hat{\phi}_{\text{Mack}} \sum_{(i,j) \in \mathcal{L}_I} |\hat{\mu}_{ij}^{\text{CL}}| = \widehat{\text{MSEP}}_{\text{Mack}}(R),$$

whenever the calibration is admissible. This equality concerns one aggregate second moment. The Mack and ODS predictive distributions can still differ, as can their quantiles. Define the informative observed-cell set

$$\mathcal{I}_I^+ := \left\{ (i, j) \in \mathcal{I}_I : j \geq 1, 0 < |\hat{\mu}_{ij}^{\text{P}}| < \infty, |X_{ij}| < \infty \right\}.$$

The Pearson-type dispersion estimator used in the numerical study is given below. Here, $|\mathcal{I}_I|$ denotes the number of observed incremental cells.

$$\hat{\phi}_{\text{Pearson}} = \frac{1}{\nu_{\text{P}}} \sum_{(i,j) \in \mathcal{I}_I^+} \frac{(|X_{ij}| - |\hat{\mu}_{ij}^{\text{P}}|)^2}{|\hat{\mu}_{ij}^{\text{P}}|}, \quad \nu_{\text{P}} = |\mathcal{I}_I| - (I + J). \quad (4)$$

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Under the square-triangle assumption $J = I - 1$, this reduces to $\nu_P = |\mathcal{I}_I| - (2I - 1)$. The expression is a nominal degrees-of-freedom correction: it treats the first development column as conditioned upon and subtracts the J selected development factors. This estimator is defined only when $\nu_P > 0$, $\mathcal{I}_I^+$ is nonempty, all required fitted quantities are finite, and the resulting calibration satisfies $\hat{\phi}_{\text{Pearson}} > 1$. The numerator uses absolute-value residuals over informative observed cells. Unavailable fitted means are omitted, including the first development column, which is conditioned upon. A zero fitted mean with a zero observed increment is omitted. A nonzero observation at a zero fitted mean makes the Pearson estimate undefined. The nominal ν_P is not reduced for such omissions, so this expression is a descriptive absolute-residual scale, not a generally unbiased ODS dispersion estimator. It is treated below as an exploratory comparator rather than a model-based Pearson estimator: an observation with the sign opposite to its fitted age sign can have a small absolute-value residual. If either calibration is at most one, non-finite, or otherwise inadmissible, it does not define the fitted negative-binomial distribution.

2.5 Implementation and model checks

Choose a monetary unit in which observed increments are integers and reconcile them with the cumulative incurred triangle. Fit the selected CL completion to $\mathcal{I}_I$ and check the factor, ultimate and variance conditions above. Use its future means to construct the fitted components and select an admissible dispersion by eqs. (3) and (4). Simulate future cells from the fitted distribution, or use the two aggregate draws in proposition 1 when only R is needed; the latter do not produce cell-level scenarios. Report $\hat{R}^{\text{CL}}$ and equal-tailed predictive intervals conditional on the fitted values.

Inspect signed and absolute observed-cell residuals by age and accident period, investigate large releases, and test sensitivity to the monetary unit, CL selection and dispersion calibration. Backtest on earlier valuations and compare with established reserving assessments where possible. Record the selections, exclusions and any failed admissibility checks. The plug-in intervals do not explicitly capture parameter or model uncertainty.

3 Structural properties

The following lemma derives the cell moments from the signed negative-binomial construction for $i = 1, \dots, I$ and $j = 0, \dots, J$, using the conditional cell mean μ_{ij} defined in section 2.

Lemma 1 (Cell moments). *Under model 1, conditional on fixed (θ, φ) , the cells are mutually independent and*

$$\begin{aligned}\mathbb{E}[X_{ij} \mid \theta, \varphi] &= \alpha_i(\beta_j^+ - \beta_j^-), \\ \text{Var}(X_{ij} \mid \theta, \varphi) &= \varphi \alpha_i(\beta_j^+ + \beta_j^-).\end{aligned}$$

When $\beta_j^+ \beta_j^- = 0$, the variance reduces to $\varphi |\mu_{ij}|$.

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Proof. Each nonzero component has shape $\alpha_i \beta_j^\pm / (\varphi - 1)$ and success probability $1/\varphi$, hence mean $\alpha_i \beta_j^\pm$ and variance $\varphi \alpha_i \beta_j^\pm$. A zero component has both moments zero. Independence of the two components gives the difference's mean and the sum of their variances. Under the one-sided restriction, $\alpha_i(\beta_j^+ + \beta_j^-) = |\mu_{ij}|$. $\square$

Proposition 1 (ODS aggregate distribution and moments). *Under assumption 2, conditional on fixed (θ, φ) , define the future component-mean totals and component sums by*

$$M_+ := \sum_{(i,j) \in \mathcal{L}_I} \alpha_i \beta_j^+, \quad M_- := \sum_{(i,j) \in \mathcal{L}_I} \alpha_i \beta_j^-, \quad R_+ := \sum_{(i,j) \in \mathcal{L}_I} Y_{ij}^+, \quad R_- := \sum_{(i,j) \in \mathcal{L}_I} Y_{ij}^-.$$

Then R_+ and R_- are independent, with

$$R_+ \sim \text{NB}\left(\frac{M_+}{\varphi - 1}, \frac{1}{\varphi}\right), \quad R_- \sim \text{NB}\left(\frac{M_-}{\varphi - 1}, \frac{1}{\varphi}\right).$$

When $M_\pm = 0$, the corresponding total is zero almost surely (including an empty future triangle). We have for the aggregate future incurred development that $R = R_+ - R_-$. Its mean and variance are

$$\mathbb{E}[R \mid \theta, \varphi] = M_+ - M_- = \sum_{(i,j) \in \mathcal{L}_I} \mu_{ij}, \quad \text{Var}(R \mid \theta, \varphi) = \varphi(M_+ + M_-).$$

Only when $\beta_j^+ \beta_j^- = 0$ at every future age does $M_+ + M_- = \sum_{(i,j) \in \mathcal{L}_I} |\mu_{ij}|$.

Proof. By ODS2, the nonzero component variables are mutually independent negative-binomial variables with common success probability $p = 1/\varphi$. A zero component is zero almost surely. Under the parametrisation used here, the probability-generating function of $N \sim \text{NB}(r, p)$ is

$$G_N(z) = \left(\frac{p}{1 - (1-p)z} \right)^r.$$

Hence, for any finite index set $\mathcal{K} = \{1, \dots, m\}$ and independent $N_k \sim \text{NB}(r_k, p)$,

$$G_{\sum_{k \in \mathcal{K}} N_k}(z) = \prod_{k \in \mathcal{K}} G_{N_k}(z) = \left(\frac{p}{1 - (1-p)z} \right)^{\sum_{k \in \mathcal{K}} r_k},$$

so that

$$\sum_{k \in \mathcal{K}} N_k \sim \text{NB}\left(\sum_{k \in \mathcal{K}} r_k, p\right).$$

For an empty set the total is zero almost surely. Otherwise its shape is the positive real sum of the component shapes. Applying the identity separately to the positive and negative components yields

$$R_+ \sim \text{NB}\left(\frac{M_+}{\varphi - 1}, \frac{1}{\varphi}\right), \quad R_- \sim \text{NB}\left(\frac{M_-}{\varphi - 1}, \frac{1}{\varphi}\right),$$

with a zero total interpreted as zero almost surely. The two totals are independent because the component collections are mutually independent. Their difference is R , and the stated moments follow from the negative-binomial parametrisation. $\square$

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The aggregate has a Skellam-like mean–variance relationship. Compared with a classical Skellam variable with component means M_+ and M_- , its mean is unchanged and its variance is multiplied by φ (Skellam, 1946). The fitted distribution has the same aggregate representation after replacing unknown parameters by fitted values. In particular, by the construction in section 2,

$$\widehat{\mathbb{E}}_I[X_{ij}] = \hat{\mu}_{ij}^{\text{CL}}, \quad (i, j) \in \mathcal{L}_I,$$

and hence

$$\widehat{\mathbb{E}}_I[R] = \hat{R}^{\text{CL}}, \quad \widehat{\text{Var}}_{I, \text{proc}}(R) = \hat{\varphi}_I \sum_{(i,j) \in \mathcal{L}_I} |\hat{\mu}_{ij}^{\text{CL}}|.$$

For the accident-year ultimate defined in section 2,

$$\widehat{\mathbb{E}}_I[U_i^{\text{ODS}}] = \hat{U}_i^{\text{CL}}, \quad i = 1, \dots, I.$$

The fitted ODS distribution preserves the selected CL mean surface and gives the aggregate process variance stated above. The aggregate representation supports two simulation methods. One simulates future cells independently and the other draws the two aggregate negative-binomial components directly. For aggregate reserve intervals, this needs at most two draws rather than one per nonzero future cell. It can therefore reduce computational work substantially, especially relative to a non-parametric bootstrap that repeatedly resamples and refits triangles. The numerical study uses the aggregate sampler for the reported results. Under Mack-MSEP calibration, the variance target includes the parameter-estimation contribution to aggregate MSEP, but the plug-in predictive draws do not reproduce its full distribution or explicitly resample fitted parameters.

Proposition 2 (ODP moment special case). *If $\beta_j^- = 0$ for $j = 0, \dots, J$, then for $i = 1, \dots, I$ and $j = 0, \dots, J$,*

$$\mathbb{E}[X_{ij} \mid \theta, \varphi] = \alpha_i \beta_j^+, \quad \text{Var}(X_{ij} \mid \theta, \varphi) = \varphi \alpha_i \beta_j^+,$$

which is the one-sided over-dispersed Poisson (ODP) mean–variance structure (England and Verrall, 2002). This is a moment relationship: the ODS component is negative-binomial, whereas an ODP quasi-likelihood specification alone does not determine a full probability distribution.

4 Numerical study

Appendix A specifies the data-generating process (DGP) in integer units, which follows the ODS assumptions. The study evaluates prediction intervals for the aggregate future incurred development $R = \sum_{(i,j) \in \mathcal{L}_I} X_{ij}$, using the observed upper triangle to fit the incurred CL mean surface. Conditional on the fixed DGP parameters, the incremental cells are independent. Note that the DGP is not a full Mack-generating model (Mack, 1993). Mack’s assumptions concern conditional moments of cumulative development, not independence of every incremental cell. The independent-cell ODS DGP does not generally satisfy those cumulative moment conditions, so Mack-MSEP serves here only as an aggregate variance-matching target. This experiment does not test its calibration under a Mack-generating process.

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All three predictive distributions use the fitted incurred CL means and differ only in the dispersion used for future cells: the oracle process-only benchmark uses the known DGP dispersion, the Mack-MSEP method uses the aggregate Mack MSEP calibration, and the Pearson method uses the observed-cell Pearson calibration. Figure 1 previews the expected cumulative triangle and corresponding CL development factors. The early negative development pattern reflects incurred reserve releases, followed by positive later development. The parameter values and cell-generation mechanism are given in Appendix A.

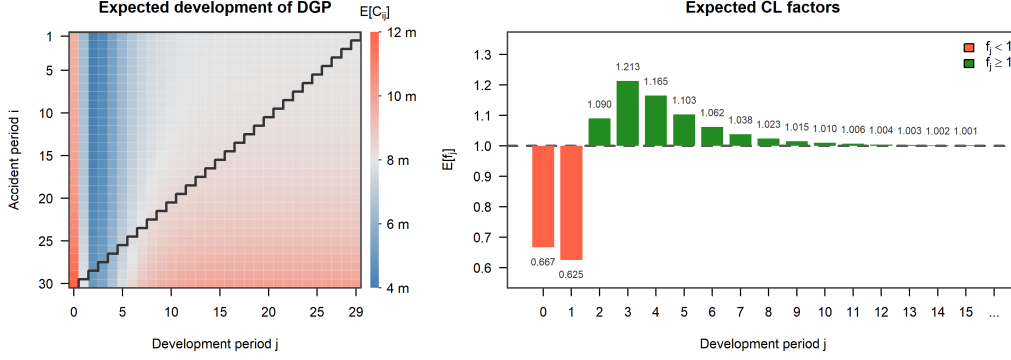


Figure 1: Overview of the simulation DGP: expected cumulative development and early CL factors with pointwise empirical 95% bands.

4.1 Simulation protocol

For each of $S = 10\,000$ outer replications, we simulate a full incremental triangle under Appendix A. We retain the observed cells in $\mathcal{I}_I$, regard the cells in $\mathcal{L}_I$ as future, and record the realised future reserve $R^{(s)}$. We then fit incurred Mack CL to the observed triangle and obtain its fitted future mean surface. The oracle process-only benchmark uses this fitted centre together with the known DGP dispersion φ_{DGP} . It is infeasible because that dispersion is not estimated from the observed triangle. The two feasible alternatives use the same fitted centre with, respectively, the Mack-MSEP calibration $\hat{\varphi}_{\text{Mack}}$ and the observed-cell Pearson calibration $\hat{\varphi}_{\text{Pearson}}$.

For each replication and for each of the three dispersion choices, we simulate $B = 10\,000$ conditionally independent reserve draws as the difference of the two aggregate negative-binomial component totals. Equal-tailed empirical intervals use probabilities (2.5%, 97.5%) and (0.5%, 99.5%) for nominal 95% and 99% coverage, respectively. Coverage is recorded when the interval contains the realised reserve $R^{(s)}$. All $S = 10\,000$ outer replications met the CL and dispersion admissibility conditions in Section 2. The reported coverage percentages therefore use all S replications for each method.

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Algorithm 1 Monte Carlo protocol for coverage assessment

Input: Simulation design in Appendix A. Set $S = 10\,000$ and $B = 10\,000$.

- 1: **for** $s = 1, \dots, S$ **do**
 - 2: Simulate a full triangle, retain $\mathcal{I}_I$, and record $R^{(s)}$ over $\mathcal{L}_I$
 - 3: Fit incurred Mack CL and obtain the fitted future mean surface
 - 4: Calculate $\hat{\phi}_{\text{Mack}}^{(s)}$ and $\hat{\phi}_{\text{Pearson}}^{(s)}$
 - 5: **for each** $\phi \in \{\varphi_{\text{DGP}}, \hat{\phi}_{\text{Mack}}^{(s)}, \hat{\phi}_{\text{Pearson}}^{(s)}\}$ **do**
 - 6: Simulate B future reserves from the fitted ODS mean surface and dispersion ϕ
 - 7: Construct 95% and 99% intervals and record coverage indicators
 - 8: **end for**
 - 9: **end for**
 - 10: Report coverage and interval-width summaries over replications
-

4.2 Results

The point-estimation summaries appear in Table 1. The DGP oracle row reports the fixed DGP expected reserve and known dispersion and is not a fitted method. The two feasible methods use the same incurred CL centre, so their average fitted reserves and average differences from the fixed DGP expected reserve are equal. The Mack-MSEP calibration matches the aggregate Mack MSEP using the component-intensity scale over $\mathcal{L}_I$. The Pearson calibration uses observed-cell residuals over $\mathcal{I}_I$ and targets a separate dispersion quantity. Neither method adds a separate parameter-draw layer.

Method	Average reserve	Difference from DGP	Dispersion used
DGP oracle	7 322 061		100 000
Mack-MSEP	7 349 952	+27 891 (0.38%)	171 995
Pearson-based	7 349 952	+27 891 (0.38%)	79 746

Table 1: Method-wise average reserve, difference from the DGP expected reserve, and dispersion used. Mack and Pearson dispersions are averages over outer replications. The DGP oracle uses the fixed DGP expected reserve and known dispersion.

All predictive methods use the same fitted incurred CL mean surface. The common average fitted reserve is 7 349 952, close to the fixed DGP expected reserve of 7 322 061, with a difference of 0.38%. The dispersion under Pearson calibration is smaller than under the two other calibrations.

Prediction-interval coverage. Figure 2 illustrates the empirical coverage and interval widths. All intervals are conditional on the fitted CL centre within an outer replication. Process-only intervals use the known DGP dispersion, so their inner draws represent future process variation but do not propagate uncertainty from estimating the dispersion or the fitted mean. Across outer replications,

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coverage reflects both the displacement of the fitted centre and future process variation. Table 2 reports empirical coverage with approximate 95% Wilson Monte Carlo intervals in brackets. These intervals quantify outer-replication uncertainty only. In this design, Mack-MSEP only slightly over-covers at 95% and is near nominal at 99%, while process-only and Pearson under-cover. These findings are specific to the stated DGP.

Method	95% PI (95% MC CI)	99% PI (95% MC CI)
Oracle process-only ($\varphi = \varphi_{\text{DGP}} = 10^5$)	88.5% [87.9%, 89.1%]	96.1% [95.7%, 96.5%]
Mack-MSEP ($\varphi = \hat{\varphi}_{\text{Mack}}$)	95.8% [95.4%, 96.2%]	99.2% [99.0%, 99.4%]
Pearson ($\varphi = \hat{\varphi}_{\text{Pearson}}$)	84.0% [83.3%, 84.7%]	93.5% [93.0%, 94.0%]

Table 2: Empirical prediction-interval coverage. Bracketed values are approximate 95% Wilson Monte Carlo intervals calculated from the displayed rates over $S = 10\,000$ outer replications.

Graphical diagnostics. Figure 2 summarises the two feasible dispersion calibrations and the 95% interval comparison across all three methods.

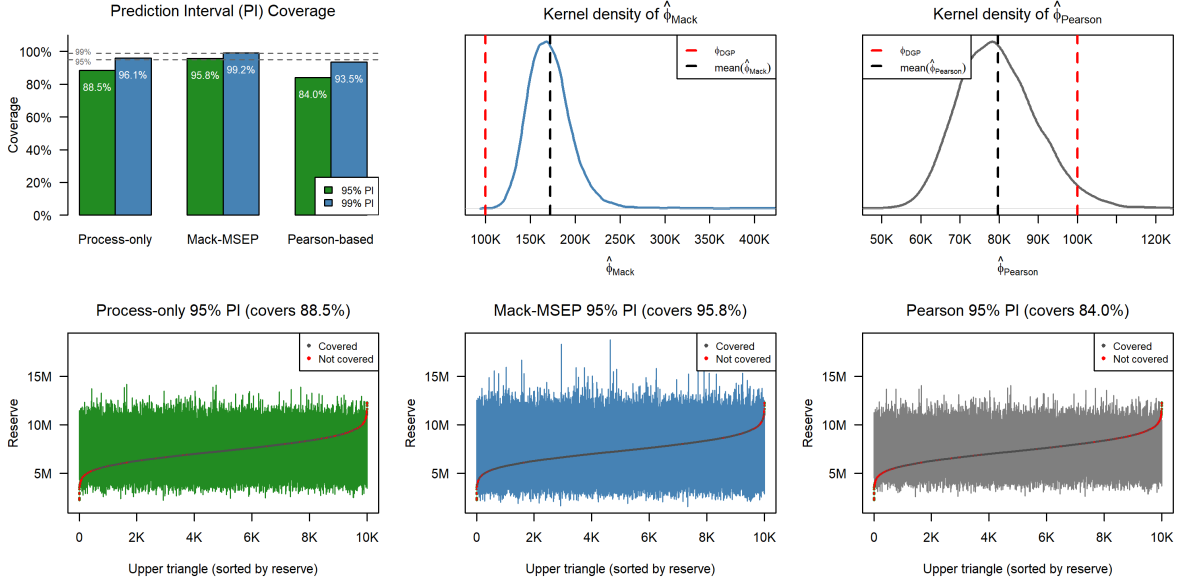


Figure 2: Distributions of the two feasible dispersion calibrations and 95% prediction intervals under process-only, Mack-MSEP, and Pearson-based dispersion.

5 Discussion

The practical role of ODS is to represent predictive uncertainty conditional on a selected incurred completion, not to select or validate the economically and legally appropriate best-estimate reserve.

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The best-estimate reserve and its decomposition into RBNS, IBNR, and future payments must be established separately. The predictive layer also requires a specified development pattern for future incurred movement around the selected completion. Its suitability depends on portfolio-specific validation of dependence, calibration, and model assumptions.

The ODS layer could also be formulated around a selected BF completion. A selected BF ultimate and development pattern imply an incremental mean surface that could replace the selected CL incremental mean surface in Section 2, subject to the stated sign and admissibility conditions. An analogous dispersion calibration could use a BF MSEP estimator (Saluz *et al.*, 2011). For comparison, a recent preprint derives an analytical MSEP for a stochastic lift of the GCC method (Richman and Wüthrich, 2026). GCC could therefore provide an alternative mean-MSEP framework for future comparison, whereas ODS provides the signed predictive distribution considered here. Numerical evaluation of the BF extension and its comparison with GCC are left to future work. The conditional formulation could apply to annual, quarterly, or monthly development periods, provided the indexing and mean selection are adapted.

Scope and limitations. The fitted one-sided sign restriction gives one active component at each future development age, although signs may differ across ages. When both signs occur among future fitted cell means, the aggregate reserve distribution has support on both sides of zero and an individual cell cannot reverse its fitted age sign. The general cell model permits both components to be active simultaneously. ODS2 in assumption 2 specifies mutual independence of all cell components, and hence cell independence conditional on the fixed model parameters. Calendar-year effects and common shocks are currently outside the present model, but other ODP/CL extensions address that separate issue (Brydon and Verrall, 2009; Wüthrich, 2010; Riegel, 2026). Under the fitted one-sided restriction, the scalar dispersion multiplier gives

$$\widehat{\text{Var}}_{I,\text{proc}}(X_{ij}) = \hat{\phi}_I |\hat{\mu}_{ij}^{\text{CL}}|, \quad (i, j) \in \mathcal{L}_I.$$

As a design choice, the analyst may assign $\kappa_j \geq 0$ to selected development periods to add predictive variance while preserving fitted means; set $\kappa_j = 0$ for periods left unchanged. Add half of κ_j to each fitted component share:

$$\tilde{\beta}_j^+ = \hat{\beta}_{jI}^+ + \frac{\kappa_j}{2}, \quad \tilde{\beta}_j^- = \hat{\beta}_{jI}^- + \frac{\kappa_j}{2}.$$

With $\tilde{\lambda}_{ij}^\pm = \hat{\alpha}_{iI} \tilde{\beta}_j^\pm$, the fitted future means remain unchanged,

$$\tilde{\mu}_{ij} = \hat{\alpha}_{iI} (\tilde{\beta}_j^+ - \tilde{\beta}_j^-) = \hat{\mu}_{ij}^{\text{ODS}}, \quad (i, j) \in \mathcal{L}_I,$$

whereas the fitted predictive variance is

$$\widehat{\text{Var}}_{I,\kappa}(X_{ij}) = \hat{\phi}_I \hat{\alpha}_{iI} \left(\hat{\beta}_{jI}^+ + \hat{\beta}_{jI}^- + \kappa_j \right), \quad (i, j) \in \mathcal{L}_I.$$

Thus, the mean is unchanged and the fitted variance increases by $\hat{\phi}_I \hat{\alpha}_{iI} \kappa_j$. How κ_j should be selected or calibrated remains for future work.

6 Conclusion

We develop ODS as a conditional predictive layer for future incurred development around a selected incurred chain-ladder completion. Whenever the fitting conditions are met, it preserves the selected CL mean in every future cell and therefore preserves the corresponding expected aggregate reserve and accident-year ultimates. The deterministic reserving centre is consequently separated from the stochastic distribution used to represent future variation.

The contribution is deliberately limited to a conditional predictive model and two scalar dispersion calibrations. Mack-MSEP calibration matches one aggregate process-variance target. It does not match the full Mack predictive distribution or propagate parameter uncertainty through explicit parameter draws. The Pearson-type calibration targets a separate retrospective quantity.

In the stated ODS-generated simulation, all methods captured the mean adequately because they shared the same fitted CL centre, while their uncertainty assessments differed: the process-only benchmark under-covered, the Mack-MSEP interval showed only slight over-coverage at 95% and near-nominal coverage at 99%, and the Pearson calibration produced narrower intervals and under-covered. These findings are specific to the stated data-generating process and motivate validation under alternative portfolios and calibration choices.

The synthetic study provides a controlled illustration of the model. Portfolio-specific validation has also been undertaken through actuarial-function work, but is outside the evidence reported here; the synthetic results should not be read as a substitute for that assessment. Further research can examine portability across portfolios, alternative dependence structures, explicit parameter uncertainty, and monetary-unit sensitivity. Mapping signed incurred development to payments or regulatory liabilities remains a separate modelling task.

A Simulation study details

We introduce the DGP used for the numerical experiment. We set $I = J + 1 = 30$ and generate a full 30×30 incremental triangle with independent cells, as in ODS1–ODS2 of assumption 2. The cell mean is

$$\mu_{ij} = \alpha_i(\beta_j^+ - \beta_j^-), \quad i = 1, \dots, 30, \quad j = 0, \dots, 29. \quad (5)$$

The ultimate amounts, expressed in an arbitrary integer monetary unit, are

$$\alpha_i = 8\,000\,000 + 2\,000\,000 \frac{\exp(3(i-1)/29) - 1}{\exp(3) - 1}, \quad i = 1, \dots, 30.$$

They increase from 8 million to 10 million. Let $\gamma_j = \beta_j^+ - \beta_j^-$. We set

$$\gamma_0 = 1.1, \quad \gamma_1 = -0.2, \quad \gamma_2 = -0.1.$$

For $k = 1, \dots, 27$, evaluate the log-normal density with log-location 1.2 and log-scale 0.6 at k :

$$w_k = \frac{1}{0.6k\sqrt{2\pi}} \exp\left[-\frac{(\log k - 1.2)^2}{2(0.6)^2}\right].$$

The remaining mass 0.2 is allocated by normalising these sampled density values:

$$\gamma_j = 0.2 \frac{w_{j-2}}{\sum_{k=1}^{27} w_k}, \quad j = 3, \dots, 29.$$

Thus $\sum_{j=0}^{29} \gamma_j = 1$. Define

$$\beta_j^+ = \max(\gamma_j, 0), \quad \beta_j^- = \max(-\gamma_j, 0).$$

Thus, development periods 1 and 2 have negative net mean increments. For $\varphi_{\text{DGP}} = 100\,000$, each cell is generated as

$$X_{ij} = Y_{ij}^+ - Y_{ij}^-, \quad Y_{ij}^\pm \sim \text{NB}\left(\frac{\alpha_i \beta_j^\pm}{\varphi_{\text{DGP}} - 1}, \frac{1}{\varphi_{\text{DGP}}}\right), \quad (6)$$

with independent components and the point-mass convention for a zero component. The value of φ_{DGP} is tied to the chosen integer monetary unit. The DGP is therefore an instance of the ODS model, with negative early incurred development and positive later development. A visual overview is provided in Figure 1 in the numerical study.

Generative AI declaration

Generative AI was used for coding assistance, writing assistance and formatting. The author takes full responsibility for the final content.

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